Check for updates

Received: 1 January 2024 | Accepted: 16 April 2025

DOI: 10.1111/1365-2745.70087

Journal of Ecology

RESEARCH ARTICLE

Ants on flowers: Protective ants impose a low but variable cost to pollination, moderated by location of extrafloral nectaries and type of flower visitor

Amanda Vieira da Silva[1] | Anselmo Nogueira[1] | Judith L. Bronstein[2] | Pedro J. Rey[3] | Laura Carolina Leal[4]

[1]Centro de Ciências Naturais e Humanas, Universidade Federal do ABC, São Bernardo do Campo, Brazil

[2]Department of Ecology and Evolutionary Biology, University of Arizona, Tucson, Arizona, USA

[3]Departamento de Biología Animal, Biología Vegetal y Ecología, Universidad de Jaén, Jaén, Spain

[4]Departamento de Ecologia e Biologia Evolutiva, Universidade Federal de São Paulo, São Paulo, Brazil

Correspondence
Amanda Vieira da Silva
Email: mandyvieira15@gmail.com

Funding information
Conselho Nacional de Desenvolvimento Científico e Tecnológico, Grant/Award Number: 434692/2018-2; Fundação de Amparo à Pesquisa do Estado de São Paulo, Grant/Award Number: 2017/13358-1, 2019/19544-7 and 2020/11171-4

Handling Editor: Ryan Phillips

Abstract

1. Throughout their lifetimes, individuals of most species engage in multiple mutualistic interactions simultaneously. These mutualisms can interfere with each other, leading to ecological costs that can influence the outcome of these interactions. Identifying factors that modulate these costs is pivotal to reaching a deeper understanding of the dynamics of mutualistic interactions.
2. We investigated the ecological costs of protective ants on the pollination of plants with extrafloral nectaries (EFNs) and how plant traits and the type of pollinator can modulate them. We hypothesized that (1) ants negatively affect the patterns of flower visitation and plant reproductive performance and that (2) this effect is relatively higher for plants bearing EFNs close to inflorescences compared to those bearing EFNs only on vegetative tissues. Additionally, we hypothesized that (3) the effect of ants on flower visitation is contingent upon the type of flower visitor. To evaluate these hypotheses, we used a meta-analytical approach based on 27 studies that assessed the effects of ants on patterns of floral visitation and plant reproductive performance.
3. We found that ants impose a low, yet highly variable, cost to pollination. Specifically, ants have a strong negative effect on patterns of flower visitation, especially when the visitors are bees, but this does not translate into reduced plant reproductive performance. In fact, the reproductive success of plants with EFNs close to vegetative tissue is enhanced by ant presence. Finally, the negative effect of ants on flower visitors is stronger when EFNs are closer to inflorescences.
4. Synthesis: Our results indicate that, in general, ants have a low impact on the pollination of EFN-bearing plants. This pattern supports theoretical predictions that natural selection acts to lower the costs of protection mutualisms. Our macroscale evaluation further shows that plant traits such as EFN location, as well as the type of flower visitors, modulate this cost. The outcome of ant–pollinator

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.
© 2025 The Author(s). *Journal of Ecology* published by John Wiley & Sons Ltd on behalf of British Ecological Society.

interactions may affect the evolutionary trajectory of traits mediating defence and pollination in EFN-bearing plants.

KEYWORDS
ant–plant interaction, ant–pollinator conflict, ecological interactions, meta-analysis, mutualism, plant defence, pollination

1 | INTRODUCTION

Individuals are often involved in multiple types of mutualisms over their lifespans. In some cases, these mutualisms are engaged in simultaneously. When this is the case, the outcome of a mutualism for the focal individual may change according to the presence of other mutualistic partners (Afkhami et al., 2014). The outcome may vary from synergistic, increasing the fitness of the focal individual (Barillas et al., 2007), to antagonistic, reducing the fitness of the focal individual to a level lower than would be expected if each mutualism were occurring separately in time (Ness, 2006). Antagonistic effects are among the ecological costs of mutualisms. They arise when partners from different mutualisms affect each other negatively or when one partner negatively affects the other, compromising the net effect of each mutualism for the focal individual (Frederickson et al., 2012; Strauss et al., 2002). Such costs exist when the same resource is provided to more than one mutualistic partner simultaneously (e.g. plants providing carbohydrates to nitrogen-fixing bacteria and to protective insects visiting extrafloral nectaries; Godschalx et al., 2015), or when one mutualistic partner repels others that confer a different mutualistic reward (e.g. protective ants repelling pollinators; Altshuler, 1999). Theoretical studies have predicted that natural selection will act to lower the costs of mutualisms (Foster & Wenseleers, 2006), and some empirical studies have shown that these costs can be low or negligible in some present-day mutualisms (e.g. Ferreira et al., 2022; Jones et al., 2017). However, other studies have reported a high cost for individuals interacting with multiple mutualists (e.g. Laird & Addicott, 2007; LeVan et al., 2014; Rico-Gray & Castro, 1996). Thus, quantifying these costs and investigating the factors modulating them is a fundamental step toward improving our understanding of the dynamics of mutualism.

In mutualistic interactions between protective ants and plants bearing extrafloral nectaries (EFNs), protective ants are attracted to extrafloral nectar, a nutritious reward produced by a secretory structure present in more than 4000 plant species worldwide (Keeler et al., 2024; Weber & Keeler, 2013). Once on the plant, they reduce herbivory by consuming or repelling herbivores (Bronstein, 1998; Frederickson et al., 2012; Rosumek et al., 2009). However, in plant species in which extrafloral nectar secretion overlaps in time with flowering, protective ants can also repel floral visitors. For example, ants can deter pollinators by behaving aggressively towards visitors or simply by being nearby (Villamil et al., 2018a). Furthermore, some visitors avoid flowers even when ants are not physically present, as flower visitors detect ant pheromones on plants (Cembrowski et al., 2014; Li et al., 2014). Therefore, the outcome of ant protective service to EFN-bearing plants should depend on both the protective benefits ants confer and the magnitude of their interference with pollination performance.

The extent of ant interference in the pollination performance of plants with EFNs is context-dependent. We focus here on two ecological factors likely to play a role in this interference. First, the spatial position of the EFNs might determine the likelihood of encounters between ants and pollinators (Hoeksema & Bruna, 2015). Extrafloral nectaries located on inflorescences or flower bracts (hereafter, EFNs on reproductive structures) can reduce herbivory on reproductive structures during their development (Oliveira et al., 1999), whereas EFNs on developing branches and leaves (hereafter, EFNs on vegetative structures) can reduce herbivory on plants' vegetative parts (Rosumek et al., 2009). Extrafloral nectaries on vegetative structures could also reduce the number of foraging ants on reproductive structures, such as open flowers (Villamil et al., 2018b; Wagner & Kay, 2002). In this case, EFNs positioned exclusively on plant vegetative parts could enhance the protection of vulnerable vegetative structures while simultaneously reducing potential conflicts between ants and pollinators by decreasing ant foraging on flowers. Conversely, plants with EFNs on reproductive structures or on both reproductive and vegetative structures (Keeler et al., 2024; Weber & Keeler, 2013) could experience more frequent ant–pollinator encounters or flower avoidance by pollinators. Thus, we expect the magnitude of ant interference on pollination to be relatively higher for plants with EFNS on reproductive than on vegetative structures.

Second, the type of flower visitors can also be an important driver of ant–pollinator outcome. Some but not all floral visitors can recognize and avoid flowers on which ants have foraged (Miner & Wilson Rankin, 2023; Rankin et al., 2018). For example, the presence of ants on *Turnera subulata* (Passifloraceae) strongly reduces the frequency of flower visitation by bees but not by butterflies (Santos & Leal, 2019), even though both floral visitors can recognize ants (Cembrowski et al., 2014; Sendoya et al., 2009). Thus, we expect that the magnitude of this avoidance depends on the type of flower visitors.

Here, we used a meta-analytical approach to quantify the overall effect of ants on the pollination of plants with EFNs. Furthermore, we explored how the location of EFNs on the plant and the type of flower visitors drive the magnitude of such effects and their consequences for the functioning of pollination mutualisms in EFN-bearing plants. We hypothesize that (1) the presence of ants has a

13652745, 2025, 8, Downloaded from https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70087 by Capes, Wiley Online Library on [04/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

negative effect on both the patterns of flower visitation and plant reproductive performance, (2) this effect is higher on plants with EFNs on reproductive structures compared to those with EFNs only on vegetative structures, and (3) the influence of ants on flower visitation is contingent upon the type of flower visitor.

2 | METHODS

2.1 | Data search

We searched for studies investigating the effects of protective ants on the pollination of EFN-bearing plants, including studies exploring flower visitation and plant reproductive performance metrics of pollination success. To accomplish this, we used a combination of search methods. First, we searched the ISI Web of Knowledge main core library and the Scopus database, considering studies published between January 1945 and September 2022. We used the following keywords: "(ant-plant OR myrmecoph*) AND (pollina* OR robbe* OR thie* OR larcen*)". The keywords from the 158 studies obtained through this search were then extracted using the *litsearchR* package v. 0.4.1 (Grames et al., 2019, 2020) in software R v. 4.0.5 (R Core Team, 2022). We used the most common keywords to create four new sets of keywords that were used in three additional searches on the same databases: (1) "(extrafloral nectar*) AND (pollina* OR robbe* OR thie* OR larcen*)", (2) "(ant-pollinator conflict*) AND (pollina* OR robbe* OR thie* OR larcen*)", and (3) "(ant-plant mutualis* OR ant-plant interact*) AND (pollina* OR robbe* OR thie* OR larcen*)". In addition, we compiled all the studies citing Altshuler (1999) and Ness (2006), the oldest and the most cited study, respectively, on the effect of ants on pollination. We also surveyed the references reported in reviews and book chapters located through our literature search ($N = 92$) to check for relevant studies that had not been found in our previous searches ($N = 34$). To be as exhaustive as possible, we also screened the literature cited within all the papers from our systematic search and selected the studies that followed our eligibility criteria (see the description below) ($N = 57$). Together, these search methods returned 1163 studies. After compiling these studies into one temporary database, we used the *litsearchR* package to identify and remove duplicates ($N = 635$). *LitsearchR* failed to identify 32 duplicates in our dataset, which we then removed manually. We excluded two additional studies (one book chapter and one original study) because we had no access to them. To broaden our data set, we included six papers in languages other than English that were found during our searches. This inclusion reduces bias in meta-analyses (Konno et al., 2020). In the end, our primary database comprised 567 studies.

2.2 | Eligibility criteria and study selection

To determine whether papers were suitable for our meta-analysis, we read all their titles and abstracts and checked whether they satisfied six eligibility criteria. First, the studies must have evaluated the role of ant attendance on any descriptor of flower visitation and/or plant reproductive performance by contrasting plants or branches with ants feeding at EFNs against plants or branches without ants. Regarding ant exclusion, studies used three different approaches to ensure that plants were not visited by ants: (1) plants lacked ants naturally ($k = 38$ effect sizes, $N = 4$ studies), (2) researchers removed ants manually, then added a sticky resin on the branch or stem to keep ants from visiting ($k = 94$, $N = 23$) or (3) researchers removed ants manually, then added a polytetrafluoroethylene-coated acetate disc around the stem ($k = 2$, $N = 1$). If a study evaluated the role of ant density or abundance on flower visitors, we included it if the authors summarized the effect of ant presence by providing an estimated central tendency value and variance for plants in which ants were present and absent (e.g. Unni et al., 2021). Second, we only considered studies with spatial and/or temporal overlap between ant attendance at EFNs and flower visitation. Thus, we did not include studies in which plants secreted nectar from EFNs only during the fruiting period, since there would be no overlap between ants and pollinators (Sanz-Veiga et al., 2017). For the same reason, we did not include studies focused on fig trees (Jander, 2015; Lin et al., 2018) because, in this system, pollinators are present on the plant only briefly before entering the inflorescence to reproduce, reducing the likelihood that they will interact with ants. We also excluded studies that experimentally evaluated the effects of ants on pollination by mimicking ant chemical cues instead of manipulating ant presence on the flowers (e.g. Ballantyne & Willmer, 2012). In these studies, pollinators did not rely on the visual cues of ant presence on flowers, which could lead to differences in the magnitude of effects reported in these studies and the studies in which the ants were visiting the flowers. Third, we did not include theoretical or modelling studies (e.g. Wang et al., 2015). Fourth, some studies evaluated the role of ants on pollination performance but did not explicitly consider the effects on flower visitation because the goal was to assess the ant effects on herbivores (Horvitz & Schemske, 1984). We included these studies only when ant exclusion occurred during flowering, with flowers noted to receive natural visits from potential pollinators during the experiment (i.e. flowers were not pollinated manually). Fifth, we did not consider studies in which ants visited honeydew-secreting insects occurring on plants with EFNs, given that honeydew attractiveness could intensify or reduce the ant effect on pollinators. Furthermore, these insects are herbivores and can directly decrease plant performance (e.g. Watanabe et al., 2018). Finally, we did not include studies investigating obligate protective mutualisms between ants and plants (i.e. myrmecophytes) because the higher dependence on ants in these highly specialized systems has likely favoured different strategies for regulating ants' effects on flowers and pollination (Chamberlain & Holland, 2009). Based on these criteria, we identified 28 studies suitable for our meta-analysis (see Figure S1 for the PRISMA flowchart describing all studies removed and the reasons, Figure S2 for a map in which studies were conducted, and Table S1 for the PRISMA ecology and evolution checklist; O'Dea et al., 2021).

13652745, 2025, 8, Downloaded from https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70087 by Capes, Wiley Online Library on [04/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Type of ant effect	Response variables
Flower visits	Frequency of visits per flower[a], frequency of visits per plant[a], frequency of floral visitors[a], duration of visits[b]
Plant reproductive performance	Number of fruits produced per plant, proportion of immature or mature fruits, number of fertilized ovules per total number of ovules produced, number of seeds per bud, number of seeds per inflorescence, number of seeds per fruit, number of seeds per plant, number of viable seeds, percentage of ovules that developed into seeds, number of plants that set seeds, seed mass

TABLE 1 Response variables retrieved from studies as proxies of ants' indirect (flower visit metrics) and direct (plant reproductive performance metrics) effects on the pollination of EFN-bearing plants.

Note: Given that the same variables may receive different names across studies, we standardized the names of the variables here. The original names are available in the meta-analytical dataset.

[a]Frequency of visits per flower or plant or frequency of floral visitors was estimated with a variable observation time across studies.

[b]Duration of visits is commonly recognized as handling time in other studies, although the authors reviewed here did not use this term; declining duration of flower visits is expected to affect pollination success negatively since the shorter the visit, the less pollen can be collected and/or deposited on the stigmas.

2.3 | Data extraction

To estimate the mean effect of ants on pollination, we built a dataset from the 28 suitable studies. These studies used a variety of response variables that are indirectly (flower visit metrics) or directly (plant reproductive performance metrics) related to pollination success (Table 1). We collected the mean, standard deviation (SD), and sample sizes of response variables from treatments with and without ant attendance from each study. When only standard errors were available, we transformed them into SD. When the data were available in figures, we used the *metaDigitise* package v. 1.0.1 (Pick et al., 2018) to extract means and SDs from each treatment ($k = 47$ effect sizes). We also used *metaDigitise* to obtain means and SDs from boxplots presenting the minimum and maximum value, lower and upper 75th quartile, and sample sizes ($k = 10$). When there were uncertainties about the information reported in the study or missing data, we contacted the authors requesting more information ($N = 9$ studies). In two studies, the authors provided the total sample size, but not for each treatment (with and without ants) and did not respond to our query; in these cases, we assumed a balanced sample size between treatments. Otherwise, we excluded the study in cases where we could not obtain the information directly from the authors ($N = 3$). One study (Lach, 2008) satisfied our inclusion criteria but evaluated the effect of ants on the entire arthropod community. In this case, we only extracted effect sizes from metrics describing the effects of ants on bee pollinators. Approximately 80% of the studies estimated the mean effect of ants on more than one metric describing flower visits (e.g. duration and the number of visitors) or plant reproductive performance (e.g. number of seeds and fruits) ($N = 22$). Consequently, more than one estimate of the ant effect size was associated with most studies. Finally, one study evaluated ants' effects on plants in different years (Del-Claro & Marquis, 2015); we extracted the effect sizes separately for each year.

2.4 | Calculating effect sizes

We used Hedges' *g* (Hedges & Olkin, 1985) as the effect size metric. We calculated Hedges' *g* from the mean, SD and sample sizes of metrics describing the effects of ants on patterns of flower visits and plant reproductive performance in each study using the *metafor* package v. 2.4.0 (Viechtbauer, 2010). We subtracted the mean values from the treatment in which ants were absent on plants from those in which the ants visited the flowers. Positive values of Hedges' *g* imply that ants positively affect pollination (e.g. there are more pollinator visits or more seeds produced in plants tended by ants). Negative values of Hedges' *g* imply that ants have a negative effect on pollination (e.g., there are fewer pollinator visits or fewer seeds produced by plants tended by ants).

2.5 | Moderating factors

In our meta-analytical models, we used combinations of three categorical moderators: the type of pollination metric, the location of EFNs on plants and the type of flower visitor (Table 2). Below, we describe how information for each moderator was extracted and how these data were assigned to different categorical levels.

2.5.1 | Pollination metrics

We classified the response variables reported in each study as metrics indirectly (flower visits) or directly (plant reproductive performance) related to pollination success (Table 1). Plant reproductive performance metrics can be separated into metrics representing ant effects on pre-pollination events (e.g. number of pollinated flowers) and metrics representing post-pollination events (e.g. fruit or seed set). Pre-pollination metrics are more closely related to patterns of flower visitation than are post-pollination metrics (Chamberlain

13652745, 2025, 8, Downloaded from https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70087 by Capes, Wiley Online Library on [04/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 Hypotheses and predictions for the effect of protective ants on pollination success driven by different moderators: The type of pollination metric used within studies, location of extrafloral nectaries, and type of flower visitor.

Moderator	Levels	Hypotheses	Predictions
Pollination metrics	Floral visitation or plant reproductive performance	Ants will negatively affect floral visitation and plant reproductive performance	A negative mean Hedges' *g* for floral visitation and plant reproductive performance
Extrafloral nectary location	Reproductive or vegetative	The negative effect of ants on pollination will be stronger for plants with EFNs on reproductive than on vegetative structures	A lower mean Hedges' *g* for plants with EFNs on reproductive than for plants with EFNs on vegetative structures
Type of flower visitor	Bees or butterflies	The magnitude of the effect of ants will depend on the type of flower visitor	Different negative mean values of Hedges' *g* for different flower visitors

Note: Levels indicate the categories evaluated for each moderator. Predictions were based on Hedges' *g* effect size estimate.

et al., 2014), which could bias the estimates of ant effects on plant reproductive performance. Since we found only five observations (metrics: pollination efficiency, the proportion of pollinated flowers, pollen load and pollen donated) from three studies that measured the effects of ants on pre-pollination events, we removed these observations from our dataset (Table 1). Therefore, ant effect sizes on plant reproductive performance are all related to effects on post-pollination events. By removing these observations, we ended up removing one study from our dataset, such that our final dataset included 27 studies and 116 effect sizes (48 of flower visitation and 67 of plant reproductive performance).

2.5.2 | Location of extrafloral nectaries

Most studies reported the location of EFNs on the plant (73%). When this information was unavailable, we retrieved it from the World Nectaries Database (Keeler et al., 2024) using the plant species name. We grouped the location of EFNs on plants into two categories: EFNs on reproductive structures (located on the base of the flower or inflorescence, bract or sepals) and EFNs on vegetative structures (located on the leaves or stipules). If the plant species bears EFNs on both reproductive and vegetative organs, we included it in the EFNs on reproductive structures category, since we were interested in ant attraction to the reproductive organs and their consequences for pollination, regardless of the presence of EFNs on leaves and branches. If the plant species bore EFNs on fruits only (pericarpial nectaries), we did not include it in our dataset. However, if the plant bore EFNs on both fruits and non-reproductive structures, we classified them as EFNs on vegetative structures (e.g. *Croton sarcopetalus*, Euphorbiaceae).

2.5.3 | Flower visitors

We collated information on the type of flower visitors reported in each study. In some cases, this varied depending on the site, season and/or year of study. For this moderator, we only included studies in which response variables were related to patterns of floral visitation (Table 1) since our focus was on the response of different visitors to ants' presence on plants. Most studies included in our dataset (80%) listed the species visiting the flowers but did not report the effects of ants on flower visits for each species or taxonomic group separately. In these cases, we were not able to estimate the effect size of ants on flower visitation by each flower visitor. For this reason, the dataset used to evaluate this specific moderator was reduced compared to the others. One study reported one effect size of ants on flower-visiting wasps (Villamil et al., 2022) and another on flower-visiting flies (Jiang et al., 2019). These effect sizes were removed from the model evaluating the effects of ants on different visitors (but kept in the others) because there was only one effect size for each flower visitor group. In contrast, there were comparatively many studies assessing the effects of ants on bees and butterflies, pollinators with contrasting sensory abilities and behaviours. Therefore, effect sizes were compared between bees and butterflies only. It is important to note that most studies evaluating the effects of ants on flower visitors did not specify whether these flower visitors were effective pollinators (i.e. whether they transferred pollen to the stigma). Therefore, we described them as flower visitors rather than pollinators. Initially, we aimed to compare the effects of ants on pollinators, nectar robbers, and nectar and pollen thieves (as one may notice in our keyword search). However, only one study reported the effect of ants on nectar thieves (Leal et al., 2006).

2.6 | Statistical analysis

To evaluate the overall mean effect of ants on flower visitation and plant reproductive performance, we built a random-effect meta-analytical model with a multilevel structure using the *metafor* package v. 2.4.0 (Viechtbauer, 2010). In this model, we used the study and plant identity as random variables to address the non-independence of data from the same study and plant species. We also considered the non-independence of plant species due to their common ancestry, including a variance–covariance phylogenetic matrix as a random variable. To do this, we built an ultrametric phylogeny comprising all

13652745, 2025, 8, Downloaded from https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70087 by Capes, Wiley Online Library on [04/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

25 plant species in our dataset using the *V.phylomaker2* package v. 0.1.0 (Figure S3; Jin & Qian, 2022). Then, we transformed the tree into a variance–covariance phylogenetic matrix using the *ape* package v. 5.6.2 (Paradis & Schliep, 2019).

To evaluate whether moderators explained variation in the effect of ants on flower visitation and plant reproductive performance, we built three random-effect multilevel meta-analytical models, one for each moderating factor (pollination metrics, EFN location, and type of flower visitor). For EFN location, we included the interaction between EFN location and the type of pollination metric in the models. For all models, we used the Hedges' g value as the response variable, weighted by the inverse of its variance and the same random factors described above, including the variance–covariance phylogenetic matrix. Our predictions are shown in Table 2.

After building our models, we observed two outlying effect sizes. The first (Hedges' $g = 17205.25$) described the effect of ants on the number of flower visits by Diptera (Jiang et al., 2019). This value may be related to how Hedges' g is calculated since, in this case, the number of visits was similar between plants with and without ants, and the SD of both groups was close to zero. Therefore, we removed this size from all analyses. The second was a negative effect size (Hedges' $g = -28.00$, approximately 933 times lower than the overall mean effect size) describing the effect of ant attendance on the reproductive success of *Cucurbita maxima* (Cucurbitaceae) (Sinu et al., 2017). In all models, the presence of this outlier did not qualitatively change the fitting or results. However, its exclusion improves data visualization and allows the reader to see the confidence interval, which the outlier would compress. For this reason, we have presented the results without it. We show the results with this outlier in the Supporting Information (Table S2). We checked the assumption of normality for all models by visually inspecting the standardized residuals of each model. All residuals were considered normally distributed. We considered significant all models in which p-values of the Q-statistics were lower than 0.05.

2.7 | Publication bias, temporal trends, and data heterogeneity

Two biases commonly affecting the fitting of meta-analytical models are (1) outcome report bias, in which significant results are easier to publish, leading to an absence of nonsignificant results, and (2) publication temporal bias, which recognizes that research methods commonly improve over time, increasing the precision of effect sizes reported and leading to nonsignificant results when results are pooled over time (Gurevitch et al., 2013; Koricheva et al., 2013). To evaluate these biases in our dataset, we built two meta-analytic models that included the same response and random variables used in the models described above (Nakagawa et al., 2022). In the first model, we included the square root of the variance as a moderator to evaluate whether studies with significant results are more likely to be published. In the second, we included as a moderator the year of publication.

Finally, to evaluate the differences in the magnitude of the effects of random variables in each model, we calculated the I^2 for each of our models (Higgins, 2003). This metric evaluates how much heterogeneity each random variable explains in each meta-analytical model. The I^2_{total} is the sum of each I^2 from each random variable. It varies from 0 to 100%; the higher its value, the higher the heterogeneity.

2.8 | Sensitivity analysis

To evaluate whether individual effect sizes affected the mean effect size estimate, we conducted a sensitivity analysis by running the overall meta-analytical model and the three models, each with one moderator, after removing each individual effect size. This procedure improves confidence in our results.

3 | RESULTS

Our dataset consisted of 116 effect sizes (k) from 27 studies (N) and 25 plant species. Nineteen studies (70.4%) reported the effect of ants on flower visitation or on plant reproductive performance only, while eight studies reported the effect of ants on both ($k = 49$). Most plant species belonged to Passifloraceae (30.2% of total effect sizes), Malpighiaceae (19.8%), and Fabaceae (10.3%). Eleven plant species bear EFNs only on vegetative tissues, while 15 bear them on reproductive or reproductive and vegetative tissues (Figure S3). One species, *Banisteriopsis malifolia* (Malpighiaceae), was reported to bear EFN on reproductive structures in one study (Baronio & DelClaro, 2018) but EFNs on vegetative structures in another (Nogueira et al., 2021).

When not taking into consideration the differences in the pollination metrics quantified in each study, we observed an overall mean neutral effect of ants on plants with EFNs (mean Hedges' $g = 0.04$, 95% CI = −0.18 to 0.26, df = 114, $k = 115$, $N = 27$). This neutral effect was associated with a large and relatively symmetric variation in the magnitude of the effect of ants, which followed a normal distribution ranging from −1.83 to 5.69. We also detected a difference in the effect of ants on flower visitation and plant reproductive performance. Ants had a negative mean effect on the patterns of flower visitation, but a positive mean effect on plant reproductive performance ($Q_{\text{pollination metric}} = 25.16$, df = 1, $k = 115$, $N = 27$, $p < 0.01$, Figure 1).

We detected an interaction between EFN location and the pollination metric type ($Q_{\text{EFN location: pollination metric}} = 26.01$, df = 3, $k = 115$, $N = 27$, $p < 0.0001$, Figure 2): the effect of ants reducing flower visitation was three times stronger in plants with EFNs on reproductive compared to plants with EFNs on vegetative structures. However, this difference did not translate into a negative effect of ants on plant reproductive performance (Figure 2). In fact, for plants bearing EFNs on vegetative structures, ants had a positive effect on plant reproductive performance (Figure 2).

13652745, 2025, 8, Downloaded from https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70087 by Capes, Wiley Online Library on [04/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

BRITISH ECOLOGICAL SOCIETY Journal of Ecology

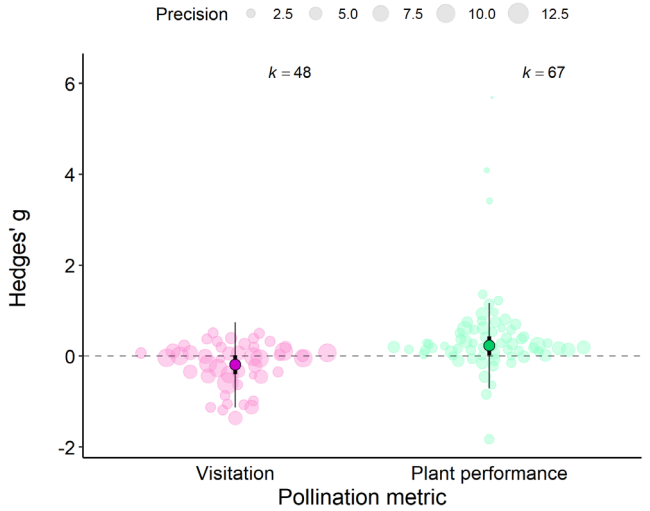


FIGURE 1 Effect of ants on flower visitation patterns and plant reproductive performance in EFN-bearing plants. Ants had a negative mean effect on flower visitation, but a positive effect on plant reproductive performance. Mean Hedges' *g* is represented by the central darker dots, 95% CI by the thicker vertical black line, and prediction interval by the thinner vertical black line. The dashed horizontal line represents no effect of ants on plants (Hedges' $g=0$). Each dot represents an effect size, *k* represents the number of effect sizes, and the size of the dot represents the precision of that effect (inverse of the standard error, larger dots represent higher precision).

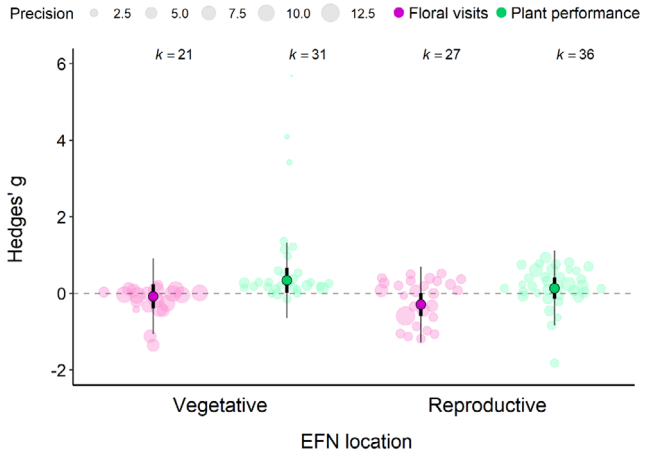


FIGURE 2 Effect of protective ants on flower visits (pink dots) and plant reproductive performance (green dots) of extrafloral nectary (EFN) bearing plant species according to the location of the EFNs. Plants bearing EFNs on reproductive structures had a threefold reduction in floral visitation compared to plants with EFNs on vegetative structures, but this difference did not translate into plant reproductive performance differences. 'Vegetative EFN' includes plants with EFNs only on vegetative structures (e.g. leaves and stipules), while 'reproductive EFN' includes plants with EFNs on both reproductive (e.g. inflorescences, bracts or sepals) and vegetative structures and those with EFNs only on reproductive structures. Mean Hedges' *g* is represented by the central circle, 95% CI by the larger vertical black line and prediction interval by the thinner vertical black line. The dashed horizontal line represents no effect of ants on plants (Hedges' $g=0$). Each dot represents an effect size, *k* represents the number of effect sizes, and the size of the dot represents the precision of that effect (inverse of the standard error, larger dots represent higher precision).

We detected that the effect of ants on patterns of floral visitation depends on the floral visitor ($Q_{\text{floral visitors}}=17.04$, df = 1, $k=41$, $N=12$, $p<0.001$, Figure 3). However, the confidence interval of the effect of ants on both bees and butterflies visitation overlaps with zero, and for bees, the overlap was minimal (ranging from −0.48 to 0.01, $p=0.06$). On average, the effect of ants on floral visitation was almost four times more negative for bees than for butterflies.

3.1 | Publication bias, temporal trends, heterogeneity and sensitivity analysis

We found a publication bias towards significant positive effect sizes for the dataset, whether the outlier (Sinu et al., 2017) was included ($QM_{\text{bias}}=34.71$, df = 1, $p<0.0001$) or excluded ($QM_{\text{bias}}=15.16$, df = 1, $p<0.0001$). Regarding temporal bias, we found a reduction of 0.02 on the estimate Hedges' *g* per year ($Q_{\text{temporal}}=3.60$, df = 1, $p=0.06$), even when we removed the outlier ($Q_{\text{temporal}}=3.73$, df = 1, $p=0.05$). Most models had a moderate level of heterogeneity, mainly explained by differences among studies (Table 3). This implies that differences among studies (e.g. population and year studied or experimental design) contributed to the variation of the effects of ants on pollination. For models with the outlier included, the heterogeneity is shown in Table S4. The sensitivity analysis showed that, after removing the outlier (Sinu et al., 2017), the sequential removal of each observation had a minimal impact on parameter estimation and did not alter the biological interpretation of the statistical model (Table S5).

4 | DISCUSSION

We evaluated the potential ecological costs of protective ants on the pollination of plants bearing EFNs by assessing the mean and distribution effects of ants on pollination, allowing us to disentangle patterns of flower visitation and plant reproductive performance. Additionally, we determined how these effects might be modulated by EFN location and the type of flower visitors. Overall, the average effect of ants on pollination was modest (near zero) but highly variable. When analysing these effects separately, we found that ants had a negative impact on flower visitation but a positive effect on plant reproductive performance. The location of EFNs influenced the extent of ant interference in flower visitation, reducing flower visits in species with EFNs on reproductive structures while having a neutral effect in plants with EFNs on vegetative structures. Complementarily, the positive effect of ants on plant reproductive performance was evident only in plants with EFNs on vegetative structures, while it remained neutral in plants

13652745, 2025, 8, Downloaded from https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70087 by Capes, Wiley Online Library on [04/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

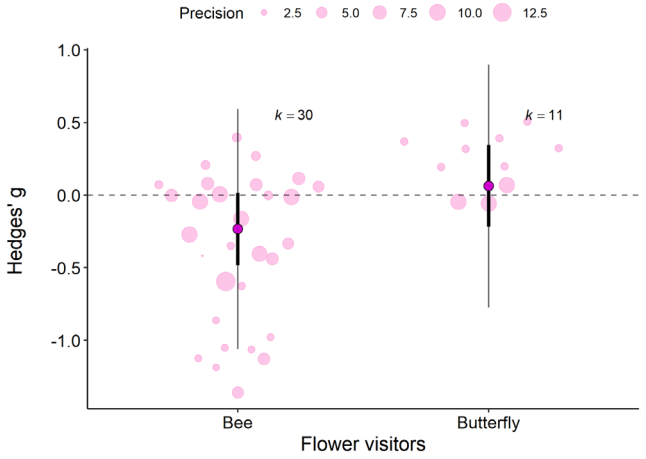


FIGURE 3 Effect of protective ants on flower visitors of EFN-bearing plant species according to the type of flower visitors. Ants had almost four times more negative effects on bees than on butterflies. Mean Hedges' *g* is represented by the central circle, 95% CI by the larger vertical black line, and prediction interval by the thinner vertical black line. The dashed horizontal line represents no effect of ants on their host plants (Hedges' $g = 0$). Each dot represents an effect size, *k* represents the number of effect sizes, and the size of the dot represents the precision of that effect (inverse of the standard error, larger dots represent higher precision).

TABLE 3 Heterogeneity estimates obtained from the models estimating the influence of the identity of the study and plant species on the ant effect on flower visitation or reproductive performance of plants bearing extrafloral nectaries (EFN).

	I^2 (95% CI)	$\sigma 2$ (95% CI)
Overall ant effect on pollination		
Study ID	86.40 (84.48–88.32)	0.30 (0.04–0.60)
Plant phylogeny	<0.01 (−1.92–1.92)	0.00 (0.00–0.61)
Plant species	<0.01 (−1.92–1.92)	0.00 (0.00–0.39)
Total	86.40 (84.48–88.32)	—
Type of pollination metric		
Study ID	82.36 (80.44–84.28)	0.22 (0.05–0.46)
Plant phylogeny	<0.01 (−1.92–1.92)	0.00 (0.00–0.26)
Plant species	<0.01 (−1.92–1.92)	0.00 (0.00–0.25)
Total	82.36 (80.44–84.28)	—
EFN location: Type of pollination metric		
Study ID	82.86 (80.93–84.78)	0.23 (0.05–0.49)
Plant phylogeny	<0.01 (−1.92–1.92)	0.00 (0.00–0.32)
Plant species	<0.01 (−1.92–1.92)	0.00 (0.00–0.25)
Total	82.86 (80.93–84.78)	—
Type of flower visitor		
Study ID	77.38 (75.46–79.30)	0.16 (0.04–0.47)
Plant phylogeny	<0.01 (−1.92–1.92)	0.00 (0.00–0.38)
Plant species	<0.01 (−1.92–1.92)	0.00 (0.00–0.17)
Total	77.38 (75.46–79.30)	—

Note: The overall model refers to the model without a moderator. The type of pollination metric considers variables describing the visitation patterns or plant reproductive performance. Type of flower visitor represents the taxonomic order of the flower visitor (bees or butterflies). Finally, EFN location means the position of EFN within the plant. Because we found an outlier (Hedges' $g = -28.00$), we reported the results of the models without the outlier here and with the outlier in Table S4.

with EFNs on reproductive structures. Furthermore, the magnitude of ant effects on flower visitation varied depending on the type of flower visitor, reducing visitation in bee-pollinated plants but not in those pollinated by butterflies. Overall, our results indicate that ants feeding on EFNs modify flower visitation patterns, but this generally does not translate into plant reproductive costs. In fact, it can even enhance plant reproductive performance in plant species with vegetative EFNs compared to when ants are absent.

One possible explanation for the intriguing pattern of the higher cost of ants on flower visitation than on plant reproductive performance could be the morphological coupling between flowers and floral visitors (e.g. Barbosa et al., 2023). For instance, bees may need to have closer contact with flowers or to stay longer to collect floral resources compared to butterflies. The hypothesis of bees being more sensitive to protective ants was proposed in a previous study, in which ants visiting the EFNs of *Turnera subulata* (Passifloraceae) negatively affected flower visitation by bees but not butterflies (Santos & Leal, 2019). Although the causes of this pattern remain to be explored, interactions with ants might increase bees' foraging costs, and they might need to visit more flowers and plants to reach their energetic requirements. For the plant, in turn, this would tend to increase outcrossing. Alternatively, since most bee species are rather generalized foragers with relatively large foraging areas (Janzen, 1971; Pasquet et al., 2008; Wikelski et al., 2010), foraging costs might be minimized by shifting to equally rewarding plant species in the same habitat. However, this behavioural shift could reduce the deposition of conspecific pollen, compromising plant reproductive performance. Although the effect of ants on pre- and post-pollination processes may differ, our dataset did not allow us to consider the effects prior to pollination. Further experiments would be needed to determine whether the foraging cost is indeed low for floral visitors, and whether and how shifts among plant species affect both foraging behaviour and plant reproductive performance, considering both pre- and post-pollination phenomena.

13652745, 2025, 8, Downloaded from https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70087 by Capes, Wiley Online Library on [04/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Several non-exclusive mechanisms could explain the low average reproductive costs of ants that we detected and their positive effect for plants bearing EFNs on vegetative structures. Although we cannot infer causation from our results, we point to some of these mechanisms here. For instance, it is possible that ant interference with flower visitation can be offset by the protective benefits that ants provide against herbivores. Interestingly, a recent meta-analysis showed that plants with EFNs on reproductive structures benefit more from anti-herbivore defence than those with EFNs on vegetative structures only (Leal et al., 2023). A balance of opposing effects of ants would be expected since the proximity of EFNs to the reproductive organs has been shown to increase the abundance of ants around these organs (Calixto et al., 2020), protecting them from florivores (Horvitz & Schemske, 1984; Oliveira, 1997). By increasing the abundance of ants around reproductive organs, it is possible that the positive effect of ants on plant reproductive performance might have been lowered due to a reduction in floral visitation. Conversely, plants bearing EFN on vegetative structures might have a lower anti-herbivory defence, but the low impact on visitation might promote a higher plant reproductive success when compared to plants bearing EFNs on reproductive structures. Therefore, reproductive EFNs could be favoured evolutionarily when flower and/or fruit predation are the main forces reducing the plant's reproductive output over time.

A second explanation for ants' relatively low effect on EFN-bearing plant performance is that ants visiting EFNs could have acted as a selective force on flower traits. One trait that could have reduced ant–pollinator conflict is related to the plant mating system. For instance, plant species with reproductive strategies less dependent on pollinators, such as self-compatible plants with higher rates of self-pollination, could pay a lower cost than plants highly dependent on pollinators for sexual reproduction. In fact, 18 plant species (72%) in our dataset were potentially self-compatible or partially self-compatible, potentially leading to reduced dependence on specific floral visitors. In addition, at a larger scale, it is possible that self-pollination might have been favoured in plants bearing EFNs on reproductive structures compared to plants bearing EFNs only on vegetative structures. However, more research is needed to evaluate the role of the mating system in reducing the ecological cost imposed by ants and how EFN location might affect plant mating systems.

Finally, a low frequency or duration of visits in the presence of ants might not represent a cost if a few or short visits are sufficient to transfer pollen among flowers. In *Turnera velutina* (Passifloraceae), ants decreased the duration of flower visits, which reduced pollen deposition and pollen export. However, because floral visitors had to visit more flowers to obtain flower resources, plants with ants experienced twice as much outcross pollen transfer, elevating pollination success (Villamil et al., 2022). A similar outcome could be expected for the reduction in the frequency of visits to a focal plant. Because bees usually visit more than one flower of a single plant during a foraging bout, they can increase geitonogamy and inbreeding depression (Karron et al., 2009). However, if bees are repelled by ants and depart to a different plant of the same species, the frequency of visits will be reduced at the focal plant, but there will be a higher chance of outcrossing. Although ants might not impose a cost to plant reproductive success even when flower visitation is reduced, we observed that plants bearing EFNs on vegetative structures suffered no effect on flower visitation and had higher reproductive success than plants bearing EFNs on reproductive structures. It is possible that the reduction in flower visits might have led to a reduction in the benefit provided by ants to plants bearing EFNs on reproductive structures.

Although ants had a low impact on plant reproductive performance, this effect was highly variable and dependent on EFN location within the plant. Because mutualistic interactions are, as a rule, highly context-dependent (Bronstein, 2015), the partners' traits and environment could help to explain this variation. For instance, aggressive dominant ants are more effective at reducing herbivory in EFN-bearing plants (Leal et al., 2023; Xu & Chen, 2010). However, they can be more aggressive against flower visitors, repelling pollinators more frequently than subordinate ants do. In addition, ants can adjust their aggressiveness towards herbivores and flower visitors according to the quality of extrafloral nectar offered by the plants and/or the availability of other food resources in the habitat (Flores-Flores et al., 2018). Likewise, plants offering a higher relative volume of extrafloral nectar and/or nectar with a higher sugar content are more attractive to protective ants (Alencar et al., 2023). In these plants, ant–pollinator encounters are more likely to occur, consequently increasing the ecological cost of ants. If plants offering extrafloral nectar on vegetative and reproductive structures simultaneously increase their chance of interacting with ants, especially aggressive ants, this would also explain a reduction in the benefit received in reproductive performance when compared to plants bearing EFNs only in vegetative structures. Although our results cannot reveal the mechanisms underlying the effect size variation within our dataset, investigating such mechanisms is a fruitful avenue for future studies.

In summary, we have shown that protective ants reduce floral visitation when EFNs are located on reproductive structures, particularly affecting bees. However, these effects did not translate into a reduction in plant reproductive performance. On the contrary, it enhanced the reproductive performance of plants bearing EFNs on vegetative structures. While our study assessed the ant–pollinator conflicts in ecological time, the outcome of this conflict could have important consequences for the evolutionary trajectory of traits mediating protective and pollination mutualism. For example, the emergence of EFNs could affect the evolution of floral phenotype or even plant mating systems, favouring traits that reduce ant–pollinator conflict. Future studies should explore whether selection to minimize ant–pollinator conflict has shaped the evolution of EFN and floral traits, especially in plant species mainly pollinized by bees, shedding new light into how ecological costs affect the evolution of mutualisms.

13652745, 2025, 8, Downloaded from https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70087 by Capes, Wiley Online Library on [04/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

AUTHOR CONTRIBUTIONS

Amanda Vieira da Silva designed the study, collected the data, performed statistical analysis, interpreted the data and wrote the first draft of the manuscript. Laura Carolina Leal and Anselmo Nogueira designed the study, interpreted the data, and revised manuscript drafts. Judith L. Bronstein and Pedro J. Rey interpreted the data and revised manuscript drafts. All authors contributed equally to the final version of the manuscript.

ACKNOWLEDGEMENTS

This study is part of Amanda Vieira da Silva's doctoral dissertation. We would like to thank Isabella Varassin, Gustavo Burin and all anonymous reviewers for their valuable suggestions. We also would like to thank all the authors who provided data for this study and answered our e-mails providing further details (Armando Aguirre Jaimes, Nico Blüthgen, Angelica E. Martinez Bauer, Nora Villamil, Nathaniel Holland and Robert Junker). Amanda Vieira da Silva thanks the São Paulo Research Foundation for the scholarship (FAPESP/grant number 2020/11171-4). Anselmo Nogueira thanks the São Paulo Research Foundation for a Young Investigators Grant (FAPESP/grant number 2019/19544-7) and the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq/grant no. 434692/2018-2). Laura C. Leal thanks the São Paulo Research Foundation (FAPESP/grant number 2017/13358-1). The Article Processing Charge for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614).

CONFLICT OF INTEREST STATEMENT

The authors declare that there are no conflicts of interest.

PEER REVIEW

The peer review history for this article is available at https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/1365-2745.70087.

DATA AVAILABILITY STATEMENT

All data and scripts used in the analysis is available in the Dryad Digital Repository (https://doi.org/10.5061/dryad.6wwpzgn9f) (Vieira da Silva et al., 2025) and GitHub (https://github.com/mandyviesi/Antsonflowers).

ORCID

Amanda Vieira da Silva https://orcid.org/0000-0002-1353-2773
Anselmo Nogueira https://orcid.org/0000-0002-8232-4636
Pedro J. Rey https://orcid.org/0000-0001-5550-0393
Laura Carolina Leal https://orcid.org/0000-0003-1570-8901

REFERENCES

*Indicates studies used in the meta-analyses.

Afkhami, M. E., Rudgers, J. A., & Stachowicz, J. J. (2014). Multiple mutualist effects: Conflict and synergy in multispecies mutualisms. *Ecology*, *95*, 833–844.

*Aguirre, A., Dáttilo, W., Rodríguez-Morales, D., Canchola-Orozco, S., Cocoletzi-Vasquez, E., Coates, R., & Angeles-Alvarez, G. (2018). Foraging ants on the extrafloral nectaries repel nectar thieves but not the effective pollinator of *Vigna luteola* (Fabaceae) in a Mexican coastal sand dune. *Sociobiology*, *65*, 621.

Alencar, C. L. D. S., Nogueira, A., Vicente, R. E., & Coutinho, Í. A. C. (2023). Plant species with larger extrafloral nectaries produce better quality nectar when needed and interact with the best ant partners. *Journal of Experimental Botany*, *74*, 4613–4627.

Altshuler, D. L. (1999). Novel interactions of non-pollinating ants with pollinators and fruit consumers in a tropical forest. *Oecologia*, *119*, 600–606.

Ballantyne, G., & Willmer, P. (2012). Floral visitors and ant scent marks: Noticed but not used? *Ecological Entomology*, *37*, 402–409.

Barillas, J. R. V., Paschke, M. W., Ralphs, M. H., & Child, R. D. (2007). White locoweed toxicity is facilitated by a fungal endophyte and nitrogen-fixing bacteria. *Ecology*, *88*, 1850–1856.

Barbosa, B. C., Delgado de Lima, T. D., Mota, G. V., & Nogueira, A. (2023). The role of intraspecific variation in bumblebee body size and behavior on buzz pollination of a tropical legume species. *American Journal of Botany*, *110*, e16236.

*Baronio, G., & Del-Claro, K. (2018). Increase in ant density promotes dual effects on bee behaviour and plant reproductive performance. *Arthropod-Plant Interactions*, *12*(2), 201–213.

Bronstein, J. L. (1998). The contribution of ant-plant protection studies to our understanding of mutualism. *Biotropica*, *30*, 150–161.

Bronstein, J. L. (2015). The study of mutualism. In J. L. Bronstein (Ed.), *Mutualism* (p. 320). Oxford University Press.

*Byk, J., & Del-Claro, K. (2010). Nectar- and pollen-gathering *Cephalotes* ants provide no protection against herbivory: A new manipulative experiment to test ant protective capabilities. *Acta Ethologica*, *13*, 33–38.

Calixto, E. S., Lange, D., Bronstein, J. L., Torezan-Silingardi, H. M., & Del-Claro, K. (2020). Optimal defense theory in an ant-plant mutualism: Extrafloral nectar as an induced defense is maximized in the most valuable plant structures. *Journal of Ecology*, *109*, 167–168.

Cembrowski, A. R., Tan, M. G., Thomson, J. D., & Frederickson, M. E. (2014). Ants and ant scent reduce bumblebee pollination of artificial flowers. *The American Naturalist*, *183*, 133–139.

Chamberlain, S., Bronstein, J., & Rudgers, J. (2014). How context-dependent are species interactions? *Ecology Letters*, *17*, 881–890.

*Chamberlain, S. A., & Holland, J. N. (2008). Density-mediated, context-dependent, consumer-resource interactions between ants and extrafloral nectar plants. *Ecology*, *89*, 1364–1374.

Chamberlain, S. A., & Holland, J. N. (2009). Quantitative synthesis of context dependency in ant–plant protection mutualisms. *Ecology*, *90*, 2384–2392.

*Del-Claro, K., & Marquis, R. J. (2015). Ant species identity has a greater effect than fire on the outcome of an ant protection system in Brazilian Cerrado. *Biotropica*, *47*, 459–467.

*do Nascimento, E. A., & Del-Claro, K. (2010). Ant visitation to extrafloral nectaries decreases herbivory and increases fruit set in *Chamaecrista debilis* (Fabaceae) in a neotropical savanna. *Flora - Morphology, Distribution, Functional Ecology of Plants*, *205*, 754–756.

Ferreira, V. C. M., Neves, F. S., & Guerra, T. J. (2022). Direct and indirect effects of ant–trophobiont interactions on the reproduction of a hummingbird-pollinated mistletoe. *Plant Ecology*, *223*, 285–296.

Flores-Flores, R. V., Aguirre, A., Anjos, D. V., Neves, F. S., Campos, R. I., & Dáttilo, W. (2018). Food source quality and ant dominance hierarchy influence the outcomes of ant-plant interactions in an arid environment. *Acta Oecologica*, *87*, 13–19.

Foster, K. R., & Wenseleers, T. (2006). A general model for the evolution of mutualisms. *Journal of Evolutionary Biology*, *19*, 1283–1293.

Frederickson, M. E., Ravenscraft, A., Miller, G. A., Hernández, L. M. A., Booth, G., & Pierce, N. E. (2012). The direct and ecological costs of an ant-plant symbiosis. *The American Naturalist*, *179*, 768–778.

13652745, 2025, 8, Downloaded from https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70087 by Capes, Wiley Online Library on [04/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

*Freitas, L., Galetto, L., Bernardello, G., & Paoli, A. A. S. (2000). Ant exclusion and reproduction of *Croton sarcopetalus* (Euphorbiaceae). *Flora*, *195*, 398–402.

Godschalx, A. L., Schadler, M., Trisel, J. A., Balkan, M. A., & Ballhorn, D. J. (2015). Ants are less attracted to the extrafloral nectar of plants with symbiotic, nitrogen-fixing rhizobia. *Ecology*, *96*, 348–354.

Grames, E. M., Stillman, A. N., Tingley, M. W., & Elphick, C. S. (2019). An automated approach to identifying search terms for systematic reviews using keyword co-occurrence networks. *Methods in Ecology and Evolution*, *10*, 1645–1654.

Grames, E. M., Stillman, A. N., Tingley, M. W., & Elphick, C. S. (2020). litsearchr: automated search term selection and search strategy for systematic reviews. R package version 0.4.1.

Gurevitch, J., Koricheva, J., & Mengersen, K. (2013). *Handbook of meta-analysis in ecology and evolution*. Princeton University Press.

Hedges, L. V., & Olkin, I. (1985). *Statistical methods for meta-analysis*. Academic Press.

*Hernandez-Cumplido, J., Forter, B., Moreira, X., Heil, M., & Benrey, B. (2016). Induced floral and extrafloral nectar production affect ant-pollinator interactions and plant fitness. *Biotropica*, *48*, 342–348.

Higgins, J. P. T. (2003). Measuring inconsistency in meta-analyses. *BMJ (Clinical Research ed.)*, *327*, 557–560.

Hoeksema, J. D., & Bruna, E. M. (2015). Context-dependent outcomes of mutualistic interactions. In J. L. Bronstein (Ed.), *Mutualism* (pp. 181–202). Oxford University Press.

*Holland, J. N., Chamberlain, S. A., & Miller, T. E. X. (2011). Consequences of ants and extrafloral nectar for a pollinating seed-consuming mutualism: Ant satiation, floral distraction or plant defense? *Oikos*, *120*, 381–388.

*Horvitz, C. C., & Schemske, D. W. (1984). Effects of ants and ant-tended herbivore on seed production of a neotropical herb. *Ecology*, *65*, 1369–1378.

Jander, K. (2015). Indirect mutualism: Ants protect fig seeds and pollen dispersers from parasites. *Ecological Entomology*, *40*, 500–510.

Janzen, D. H. (1971). Euglossine bees as long-distance pollinators of tropical plants. *Science*, *171*, 203–205.

*Jiang, X., Deng, H.-N., Qu, J., Zeng, H., & Cao, G.-X. (2019). The function of extrafloral nectaries in the pollination and reproduction of *Sambucus javanica*. *Plant Species Biology*, *34*, 53–60.

Jin, Y., & Qian, H. (2022). V.PhyloMaker2: An updated and enlarged R package that can generate very large phylogenies for vascular plants. *Plant Diversity*, *44*, 335–339.

*Jones, I. M., Koptur, S., & Peña, J. E. (2017). Exploring whether and how ants (Hymenoptera: Formicidae) affect reproductive fitness in *Senna mexicana* var. *chapmanii* (Fabaceae). *Florida Entomologist*, *100*, 539–545.

*Junker, R., Chung, A. Y. C., & Blüthgen, N. (2007). Interaction between flowers, ants and pollinators: Additional evidence for floral repellence against ants. *Ecological Research*, *22*, 665–670.

Karron, J. D., Holmquist, K. G., Flanagan, R. J., & Mitchell, R. J. (2009). Pollinator visitation patterns strongly influence among-flower variation in selfing rate. *Annals of Botany*, *103*, 1379–1383.

Keeler, K. H., Porturas, L. D., & Weber, M. G. (2024). World list of plants with extrafloral nectaries. www.extrafloralnectaries.org

Konno, K., Akasaka, M., Koshida, C., Katayama, N., Osada, N., Spake, R., & Amano, T. (2020). Ignoring non-English-language studies may bias ecological meta-analyses. *Ecology and Evolution*, *10*, 6373–6384.

Koricheva, J., Jennions, M. D., & Lau, J. (2013). Temporal trends in effect sizes: causes, detection and implications. In J. Koricheva, J. Gurevitch, & K. Mengersen (Eds.), *Handbook of meta-analysis in ecology and evolution* (pp. 237–254). Princeton University Press.

*Lach, L. (2008). Argentine ants displace floral arthropods in a biodiversity hotspot. *Diversity and Distributions*, *14*, 281–290.

Laird, R. A., & Addicott, J. F. (2007). Arbuscular mycorrhizal fungi reduce the construction of extrafloral nectaries in *Vicia faba*. *Oecologia*, *152*, 541–551.

Leal, I. R., Fischer, E., Kost, C., Tabarelli, M., & Wirth, R. (2006). Ant protection against herbivores and nectar thieves in *Passiflora coccinea* flowers. *Ecoscience*, *13*, 431–438.

Leal, L. C., Nogueira, A., & Peixoto, P. E. C. (2023). Which traits optimize plant benefits? Meta-analysis on the effect of partner traits on the outcome of an ant-plant protective mutualism. *Journal of Ecology*, *111*, 263–275.

LeVan, K. E., Hung, K. L., McCann, K. R., Ludka, J. T., & Holway, D. A. (2014). Floral visitation by the Argentine ant reduces pollinator visitation and seed set in the coast barrel cactus, *Ferocactus viridescens*. *Oecologia*, *174*, 163–171.

Li, J., Wang, Z., Tan, K., Qu, Y., & Nieh, J. C. (2014). Giant Asian honeybees use olfactory eavesdropping to detect and avoid ant predators. *Animal Behaviour*, *97*, 69–76.

Lin, S.-Y., Chou, L.-S., & Bain, A. (2018). Sugar secretion and ant protection in *Ficus benguetensis*: Toward a general trend of fig–ant interactions. *Acta Oecologica*, *90*, 168–172.

*Martínez-Bauer, A. E., Martínez, G. C., Murphy, D. J., & Burd, M. (2015). Multitasking in a plant–ant interaction: How does *Acacia myrtifolia* manage both ants and pollinators? *Oecologia*, *178*, 461–471.

*Martins, J., Carneiro, A., Souza, L., & Almeida-Cortez, J. (2020). How pollinator visits are affected by flower damage and ants presence in *Ipomoea carnea* subs. *fistulosa* (Martius and Choise) (Convolvulaceae)? *Brazilian Journal of Biology*, *80*, 47–56.

Miner, M. C., & Wilson Rankin, E. E. (2023). Bumble bee avoidance of Argentine ants and associated chemical cues. *Journal of Insect Behavior*, *36*, 20–32.

Nakagawa, S., Lagisz, M., Jennions, M. D., Koricheva, J., Noble, D. W. A., Parker, T. H., Sánchez-Tójar, A., Yang, Y., & O'Dea, R. E. (2022). Methods for testing publication bias in ecological and evolutionary meta-analyses. *Methods in Ecology and Evolution*, *13*, 4–21.

Ness, J. H. (2006). A mutualism's indirect costs: The most aggressive plant bodyguards also deter pollinators. *Oikos*, *113*, 506–514.

*Nogueira, R. R., Santos, D. F. B., Calixto, E. S., Torezan-Silingardi, H. M., & Del-Claro, K. (2021). Negative effects of ant-plant interaction on pollination: Costs of a mutualism. *Sociobiology*, *68*, e7259.

O'Dea, R. E., Lagisz, M., Jennions, M. D., Koricheva, J., Noble, D. W. A., Parker, T. H., Gurevitch, J., Page, M. J., Stewart, G., Moher, D., & Nakagawa, S. (2021). Preferred reporting items for systematic reviews and meta-analyses in ecology and evolutionary biology: A PRISMA extension. *Biological Reviews*, *96*, 1695–1722.

*Oliveira, P. S. (1997). The ecological function of extrafloral nectaries: Herbivore deterrence by visiting ants and reproductive output in *Caryocar brasiliense* (Caryocaraceae). *Functional Ecology*, *11*, 323–330.

Oliveira, P. S., Rico-Gray, V., Díaz-Castelazo, C., & Castillo-Guevara, C. (1999). Interaction between ants, extrafloral nectaries and insect herbivores in Neotropical coastal sand dunes: Herbivore deterrence by visiting ants increases fruit set in *Opuntia stricta* (Cactaceae). *Functional Ecology*, *13*, 623–631.

Paradis, E., & Schliep, K. (2019). ape 5.0: An environment for modern phylogenetics and evolutionary analyses in R. *Bioinformatics*, *35*, 526–528.

Pasquet, R. S., Peltier, A., Hufford, M. B., Oudin, E., Saulnier, J., Paul, L., Knudsen, J. T., Herren, H. R., & Gepts, P. (2008). Long-distance pollen flow assessment through evaluation of pollinator foraging range suggests transgene escape distances. *Proceedings of the National Academy of Sciences of the United States of America*, *105*, 13456–13461.

*Pereira, M. F., & Trigo, J. R. (2013). Ants have a negative rather than a positive effect on extrafloral nectaried *Crotalaria pallida* performance. *Acta Oecologica*, *51*, 49–53.

Pick, J. L., Nakagawa, S., & Noble, D. W. A. (2018). Reproducible, flexible and high-throughput data extraction from primary literature: The metaDigitise r package. *Methods in Ecology and Evolution*, *10*, 426–431.

13652745, 2025, 8, Downloaded from https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70087 by Capes, Wiley Online Library on [04/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

R Core Team. (2022). *R: A Language and environment for statistical computing*. R Foundation for Statistical Computing. https://www.R-project.org/

Rankin, D. T., Clark, C. J., & Wilson Rankin, E. E. (2018). Hummingbirds use taste and touch to discriminate against nectar resources that contain Argentine ants. *Behavioral Ecology and Sociobiology*, *72*, 44.

Rico-Gray, V., & Castro, G. (1996). Effect of an ant-aphid interaction on the reproductive fitness of *Paullinia fuscecens* (Sapindaceae). *The Southwestern Naturalist*, *41*, 434–440.

Rosumek, F. B., Silveira, F. A. O., Neves, F. d. S., Barbosa, N. P. d. U., Diniz, L., Oki, Y., Pezzini, F., Fernandes, G. W., & Cornelissen, T. (2009). Ants on plants: A meta-analysis of the role of ants as plant biotic defenses. *Oecologia*, *160*, 537–549.

*Santos, A. T. F., & Leal, L. C. (2019). My plant, my rules: Bodyguard ants of plants with extrafloral nectaries affect patterns of pollinator visits but not pollination success. *Biological Journal of the Linnean Society*, *126*, 158–167.

Sanz-Veiga, P. A., Ré Jorge, L., Benitez-Vieyra, S., & Amorim, F. W. (2017). Pericarpial nectary-visiting ants do not provide fruit protection against pre-dispersal seed predators regardless of ant species composition and resource availability. *PLoS One*, *12*, e0188445.

*Schürch, S., Pfunder, M., & Roy, A. (2000). Effects of ants on the reproductive success of *Euphorbia cyparissias* and associated pathogenic rust fungi. *Oikos*, *88*, 6–12.

Sendoya, S. F., Freitas, A. V. L., & Oliveira, P. S. (2009). Egg-laying butterflies distinguish predaceous ants by sight. *The American Naturalist*, *174*, 134–140.

*Sinu, P. A., Sibisha, V. C., Nikhila Reshmi, M. V., Reshmi, K. S., Jasna, T. V., Aswathi, K., & Megha, P. P. (2017). Invasive ant (*Anoplolepis gracilipes*) disrupts pollination in pumpkin. *Biological Invasions*, *19*, 2599–2607.

*Sobrinho, T. G., Schoereder, J. H., Rodrigues, L. R., & Collevatti, R. G. (2002). Ant visitation (Hymenoptera: Formicidae) to extrafloral nectaries increases seed set and seed viability in the tropical weed *Triumpheta semitriloba*. *Sociobiology*, *39*, 353–372.

*Stefani, V., Pires, T. L., Torezan-Silingardi, H. M., & Del-Claro, K. (2015). Beneficial effects of ants and spiders on the reproductive value of *Eriotheca gracilipes* (Malvaceae) in a tropical savanna. *PLoS One*, *10*, e0131843.

Strauss, S. Y., Rudgers, J. A., Lau, J. A., & Irwin, R. E. (2002). Direct and ecological costs of resistance to herbivory. *Trends in Ecology & Evolution*, *17*, 278–285.

*Unni, A. P., Mir, S. H., Rajesh, T. P., Ballullaya, U. P., Jose, T., & Sinu, P. A. (2021). Native and invasive ants affect floral visits of pollinating honey bees in pumpkin flowers (*Cucurbita maxima*). *Scientific Reports*, *11*, 4781.

Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. *Journal of Statistical Software*, *36*, 1–48.

Vieira da Silva, A., Nogueira, A., Bronstein, J. L., Rey, P. J., & Leal, L. C. (2025). Data from: Ants on flowers: Protective ants impose a low but variable cost to pollination, moderated by location of extrafloral nectaries and type of flower visitors [Dataset]. *Dryad Digital Repository*. https://doi.org/10.5061/dryad.6wwpzgn9f

*Villamil, N., Boege, K., & Stone, G. N. (2018a). Ant-pollinator conflict results in pollinator deterrence but no nectar trade-offs. *Frontiers in Plant Science*, *9*, 1093.

Villamil, N., Boege, K., & Stone, G. N. (2018b). Testing the distraction hypothesis: Do extrafloral nectaries reduce ant-pollinator conflict? *Journal of Ecology*, *107*, 1377–1391.

*Villamil, N., Boege, K., & Stone, G. N. (2022). Defensive mutualists affect outcross pollen transfer and male fitness in their host plant. *Oikos*, *9*, e08788.

Wagner, D., & Kay, A. (2002). Do extrafloral nectaries distract ants from visiting flowers? An experimental test of an overlooked hypothesis. *Evolutionary Ecology Research*, *4*, 293–305.

Wang, Y., DeAngelis, D. L., & Nathaniel Holland, J. (2015). Dynamics of an ant–plant-pollinator model. *Communications in Nonlinear Science and Numerical Simulation*, *20*, 950–964.

Watanabe, S., Murakami, Y., & Hasegawa, E. (2018). Effects of aphid parasitism on host plant fitness in an aphid-host relationship. *PLoS One*, *13*, e0202411.

Weber, M. G., & Keeler, K. H. (2013). The phylogenetic distribution of extrafloral nectaries in plants. *Annals of Botany*, *111*, 1251–1261.

Wikelski, M., Moxley, J., Eaton-Mordas, A., López-Uribe, M. M., Holland, R., Moskowitz, D., Roubik, D. W., & Kays, R. (2010). Large-range movements of neotropical orchid bees observed via radio telemetry. *PLoS One*, *5*, e10738.

Xu, F. F., & Chen, J. (2010). Competition hierarchy and plant defense in a guild of ants on tropical *Passiflora*. *Insectes Sociaux*, *57*, 343–349.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1. PRISMA flowchart reporting the number of studies found in our searches, the number of studies excluded in each step, and the reasons for the exclusion.

Figure S2. Studies evaluating the effect of ants on pollination included in our dataset.

Figure S3. Plant phylogenetic relationship of all species included in our dataset.

Table S1. PRISMA checklist for meta-analysis in ecology and evolutionary biology (O'Dea et al, 2021).

Table S2. List of the 27 studies included in our meta-analysis with their respective plant species, pollination metric (flower visitation and/or plant reproductive performance), list of flower visitors, and the description of the studies that reported the effect of ants separated by each floral visitor or groups of floral visitors.

Table S3. Results of all models including the outlier (Hedges' $g = -28$).

Table S4. Heterogeneity estimates obtained from the models estimating the influence of the identity of the study and plant species on the ant effect on attendance on flower visitors or reproductive success of plants bearing extrafloral nectaries (EFN).

Table S5. Sensitivity analysis to test whether any individual observation affects the mean ant effect on flower visitors or reproductive success of plants bearing extrafloral nectaries (EFN).

How to cite this article: Vieira da Silva, A., Nogueira, A., Bronstein, J. L., Rey, P. J., & Leal, L. C. (2025). Ants on flowers: Protective ants impose a low but variable cost to pollination, moderated by location of extrafloral nectaries and type of flower visitor. *Journal of Ecology*, *113*, 2187–2198. https://doi.org/10.1111/1365-2745.70087

13652745, 2025, 8, Downloaded from https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70087 by Capes, Wiley Online Library on [04/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License